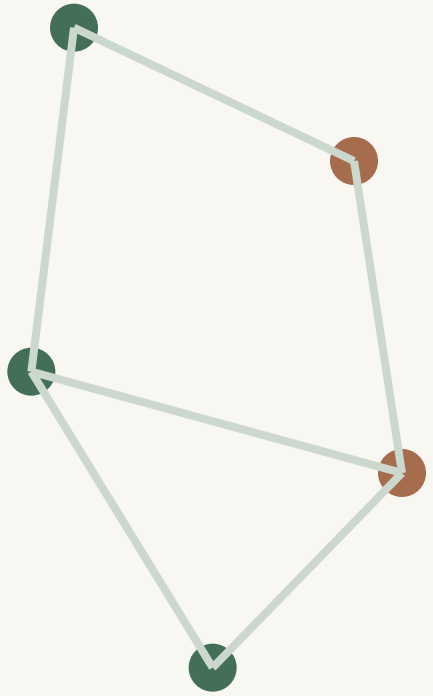
RESEARCH BRIEFING

Open-World Network-Flow Triage

A beginner-friendly walkthrough of a fixed-budget attack detection benchmark.



Explainable Cyber Threat Defense Workbench

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OVERVIEW

What this project asks

Can a limited review queue find attacks from a category left out of training?

A model trained only on known attack categories may rank those attacks well. A second model can look for unusual flows even when their attack category was not in training.

10%
of test flows can be reviewed

3
different ranking policies

9
held-out attack categories

FOUNDATION

What is a network flow?

A flow is a summary of a network connection, not the full packet contents.

One connection →

HOW LONG?

duration

HOW MUCH?

packets and bytes

WHAT TYPE?

protocol and service

UNSW-NB15 provides one row per flow with measurements such as duration, sent and received packet counts, and connection type.

The notebook ranks these rows for review. One row is not the same as a distinct incident or investigation case.

The data used in the experiment

UNSW-NB15 includes normal traffic and nine simulated attack types from a cyber range. [1]

175,341
published training flows

82,332
published test flows

73,791
primary test after overlap removal

In the primary test, 38,263 flows are attack-labeled.

That is 51.9% of the test population. This unusually high share matters when interpreting precision.

Source: UNSW-NB15 dataset [1] and executed project notebook.

FOUNDATION

What does "unseen" mean here?

The experiment creates a controlled version of an unfamiliar attack category.

DURING TRAINING

One attack category is removed. The model learns from normal flows and the other attack categories.

→

DURING EVALUATION

The withheld category stays in the test set. We ask whether its flows enter the review queue.

Example: if DoS is withheld, the known-risk model never sees a DoS-labeled training flow. We then score the test flows and count how many DoS flows appear in the queue.

This is a benchmark proxy for unknown behavior. It is not a test of real zero-day attacks.

PATTERN EVOLUTION

How detection patterns built on each other

These approaches overlap in real systems. The sequence shows how the questions expanded.

1 RULES

Does traffic match a known condition?

→

2 ANOMALY

Does traffic differ from a normal baseline?

→

3 SUPERVISED MODEL

Does it resemble labeled attacks?

→

4 OPEN-WORLD TEST

What happens when a label was never seen?

Rules, anomaly detection, and supervised scoring are established patterns. Open-world evaluation asks how those systems behave outside the known label set. [2, 3, 4]

PATTERN EVOLUTION

Examples already used in industry

These examples show existing detection patterns; they are not the project's claimed invention.

ELASTIC SECURITY [2]

Its detection engine supports rules and machine-learning anomaly alerts. Different rule types can serve different detection needs.

AMAZON GUARDDUTY

It analyzes sources including VPC flow logs and uses threat intelligence and machine learning to identify suspicious activity.

Our notebook studies one narrower question: how to divide a fixed number of review slots between a known-risk ranking and a novelty ranking.

RESEARCH CONTEXT

The open-world question is active research

Published work studies unknown-class recognition, analyst inspection, and learning as traffic changes.

OPEN RECOGNITION AND INSPECTION [3]

ORI studies how to recognize unknown samples and reduce the work needed to inspect them.

CONTINUAL LEARNING [4]

SPIDER studies network intrusion detection when data distributions shift over time.

This project contributes a transparent fixed-budget comparison on one public dataset. It does not claim a new model architecture.

SYSTEM DESIGN

Two scores answer different questions

Neither score is a verdict; each provides a ranking for review.

KNOWN-RISK SCORE

XGBoost learns from normal flows and the attack categories still present in training.

Higher score: more similar to labeled attack patterns.

NOVELTY SCORE

Isolation Forest learns from normal training flows only.

Higher score: more unusual than that normal reference.

The scores use different scales. The queue compares rankings from each lane, not their raw score values.

METHOD

One fold, using DoS as the example

The same procedure is repeated once for each of the nine attack categories.

01 **Remove DoS from training**

Normal and eight other attack categories remain.

02 **Fit both scoring models**

Known risk uses remaining labels; novelty uses normal flows.

03 **Score the test set**

DoS examples are still present in the held-out test.

The test category is used to calculate results only after the models rank flows.

METHOD

A check against exact input overlap

Identical feature rows appear on both published sides of the dataset split.

82,332
published test rows

→

- 8,541
rows matching training inputs

→

73,791
primary test

The notebook hashes the input columns, then removes test rows whose exact input values occur in the training file. Every held-out-category fold uses the same resulting test population.

This protects against exact duplicate leakage. Related flows may still share patterns.

METHOD

What information do the models see?

The main model uses 20 input columns. Labels and category names are kept out of its inputs.

12 TRAFFIC MEASUREMENTS

Duration, packets, bytes, rates, and time-to-live values.

Examples: dur, spkts, dbytes

5 RECENT-CONNECTION COUNTS

How often related service, host, and port combinations appear.

Example: ct_srv_src

3 CONNECTION CATEGORIES

Protocol, service, and connection state.

Examples: proto, service, state

Excluded: label (normal or attack), attack_cat (the held-out category), and id (row identifier).

METHOD

A review budget means a fixed-size queue

At the main setting, each policy selects the same number of test-flow rows.

73,791	x	10%	=	7,379
primary test flows		share reviewed		rows in each queue

A larger budget reviews more flows. To compare policies fairly, the notebook holds queue size equal at 5%, 10%, and 20%.

A selected row is an alert for this benchmark. It is not necessarily a distinct case for an analyst.

METHOD

Three ways to fill 7,379 review slots

The 75/25 split is a clear example policy. It was not optimized on the test labels.

KNOWN-RISK ONLY

7,379 known-risk slots

0 novelty slots

NOVELTY ONLY

0 known-risk slots

7,379 novelty slots

TWO-LANE 75/25

5,534 known-risk slots

1,845 novelty slots

For the two-lane queue, novelty chooses from flows not already selected by known-risk ranking.

Four measures, in plain language

Every measure is computed after the fixed-size queues have been selected.

HELD-OUT RECALL

Of 100 withheld-category attacks, about how many enter the queue?

KNOWN-CATEGORY RECALL

Of 100 attacks from categories used in training, how many enter?

PRECISION

Of 100 queued rows, how many have attack labels?

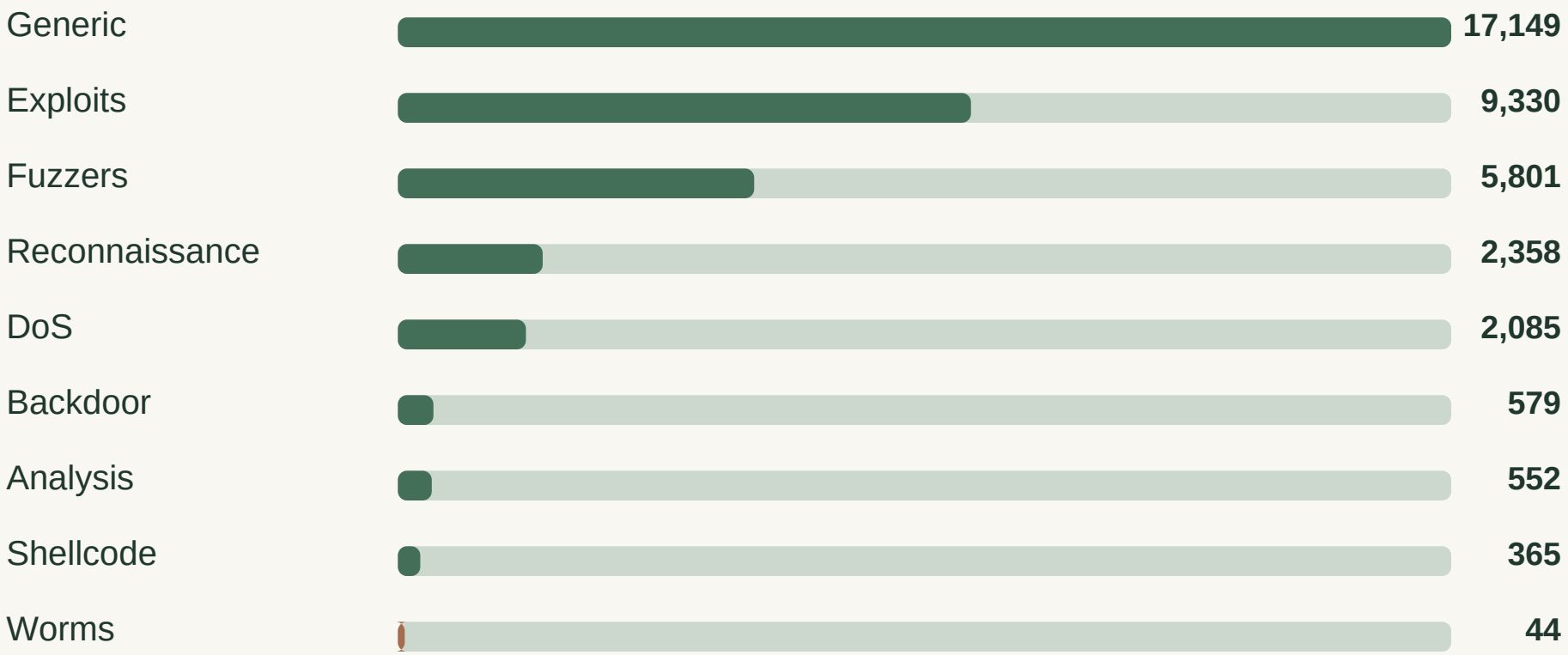
NORMAL ALERTS PER 1,000

Of 1,000 normal flows, how many take review slots?

Reported results are averages across eight categories with at least 100 test examples. Each category has equal weight.

Not every held-out category has equal support

The macro summary uses eight categories. Worms has only 44 primary-test examples.



WORMS

44

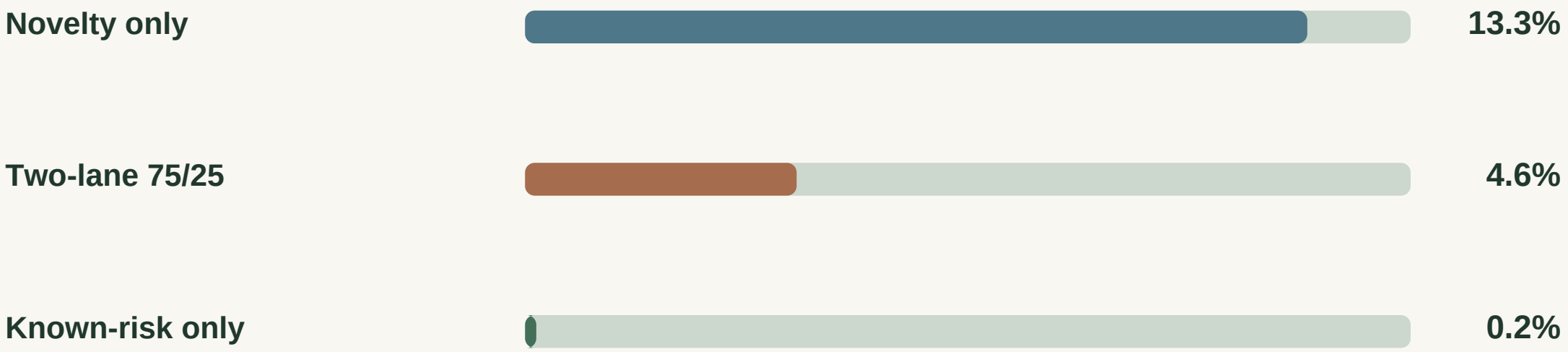
Shown separately

Small categories produce less stable recall estimates. The notebook shows each category, not just the average.

RESULTS AT 10%

Which queue finds withheld-category attacks?

Average held-out recall across eight supported categories; each queue reviews 7,379 flows.

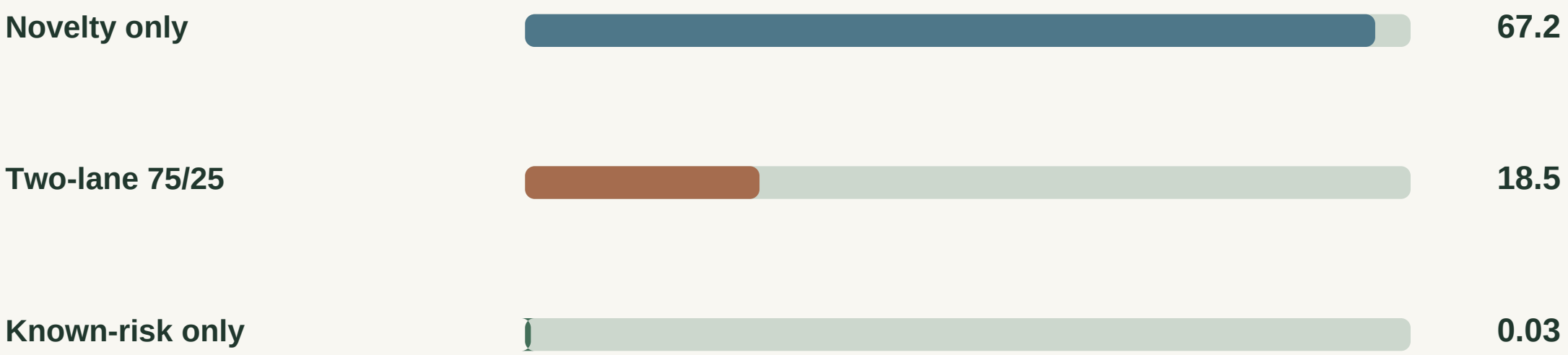


Read this as: in an average held-out category, the novelty queue selects about 13 of every 100 attack flows. The two-lane queue selects about 5.

RESULTS AT 10%

What does that cost in normal-flow review?

Average normal alerts for every 1,000 normal test flows; lower means less benign review.



The two-lane queue cuts normal-flow review by about 72% compared with novelty-only in this benchmark, while also finding fewer held-out attacks.

RESULTS AT 10%

How do the policies handle seen attacks?

The same eight-category macro summary; all policies use the same queue size.

POLICY	KNOWN-CATEGORY RECALL	PRECISION
Known-risk only	22.7%	99.98%
Two-lane 75/25	20.1%	91.1%
Novelty only	13.1%	67.6%

Precision is especially high here because more than half the test flows are attack-labeled. It is not a forecast for a live network.

RESULTS ACROSS BUDGETS

More review capacity changes the trade-off

Held-out recall is the percentage of withheld-category attacks selected into the queue.

BUDGET	KNOWN RISK	TWO-LANE	NOVELTY
5%	0.1%	2.5%	7.2%
10%	0.2%	4.6%	13.3%
20%	0.4%	8.1%	41.1%

The novelty queue catches more withheld attacks as the budget grows. Its normal alerts also rise: 37.7, 67.2, and 102.1 per 1,000 at the three budgets.

RESULTS BY CATEGORY

The average hides category differences

At the 10% budget, the two example categories below had different recall.

DOS

2,085 test flows

Novelty only	21.0%
Two-lane	11.2%

RECONNAISSANCE

2,358 test flows

Novelty only	6.7%
Two-lane	1.4%

Worms has only 44 primary-test examples. Its 9 novelty-only hits and 2 two-lane hits are reported separately, not used in the eight-category average.

INTERPRETATION

What one flow explanation shows

One held-out Exploits row illustrates how the known-risk model uses features.

0.8452

XGBoost ranking score for this flow. It is not a calibrated probability of a real-world attack.

WHAT PUSHED THE SCORE UP

Source time-to-live (sttl) and a recent source/service connection count (ct_srv_src).

WHAT PUSHED IT DOWN

The DNS service indicator and source byte count (sbytes).

These are local XGBoost SHAP contributions on a log-odds scale. They explain this model output, not why the attack happened.

CONCLUSION

What the benchmark supports

The result is a measurable policy trade-off on UNSW-NB15.

- Novelty ranking finds more withheld-category flows and sends more normal flows for review.
- A 75/25 queue recovers much of the known-category recall with less benign review than novelty-only.
- Known-risk ranking alone has almost no held-out-category recall at a 10% queue.

LIMITATIONS

What this experiment cannot establish

The benchmark answers a controlled research question on one published dataset.

- A withheld dataset category is only a proxy for unknown behavior, not a real zero-day attack.
- The CSV files have no timestamps, so they cannot show future-period or temporal-drift performance.
- The cyber-range test is attack-heavy; its precision and alert rates do not predict a live SOC queue.
- Removing exact input matches does not remove every relationship among network flows.
- Worms has only 44 test flows; there is no analyst feedback or live response in this notebook.

SOURCES

Where the data and context come from

Dataset facts and measured results come from the project notebook; selected context sources are cited below.

[1] UNSW-NB15 dataset, UNSW Research

research.unsw.edu.au/projects/unsw-nb15-dataset

[2] Elastic Security detections and alerts

www.elastic.co/docs/solutions/security/detect-and-alert

[3] Open Recognition and Inspection, IEEE TIFS 2025

ieeexplore.ieee.org/document/11159092/

[4] SPIDER, IEEE INFOCOM 2024

doi.org/10.1109/INFOCOM52122.2024.10621428